Project Proposal (Proposed Solution)

F1 Racing Analytics — interactive motorsport performance exploration

Date	15 March 2024	**Team ID**	
Project Name	F1 Racing Analytics	**Maximum Marks**	3 Marks

Project Overview

Objective

To build a clear, interactive analytics experience that helps users explore Formula 1 driver points, team performance, race outcomes and related patterns through Tableau visualisations embedded in a website.

Scope

The project includes a Bootstrap-based website with Home and Overview sections, a Tableau Public dashboard and a driver leaderboard story. The analytics cover competing teams, seasons, race outcomes, circuits, driver points, team/championship performance, tyre compounds, country participation, podiums and DNF reasons. The project is limited to the fields and records present in the supplied dataset.

Problem Statement

Description

F1 performance information can be difficult to interpret when users must inspect raw structured files or move between separate statistics and charts. Users need a consolidated view that makes driver/team comparisons and race-related patterns easier to explore.

Impact

A central dashboard can reduce the effort required to locate and compare metrics, improve the clarity of project demonstrations, and help viewers discuss performance using visual evidence. These are intended benefits rather than measured outcomes.

Proposed Solution

Approach

Connect the supplied F1 dataset files to Tableau, create the required calculations and visualisations, publish the dashboard and driver story to Tableau Public, and embed both views in a responsive website. Validate the layout, navigation, and available Tableau interactions across desktop and smaller screens.

Key Features

• KPI overview for competing teams, seasons, logged race outcomes and circuits.
• Driver points leaderboard with a Top N control in the story.
• Team performance ranking and championship-by-season analysis.
• Tyre compound, country participation, podium and DNF analysis.
• F1-themed home page, responsive navigation and embedded interactive visualisations.

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware — Computing	Computer used to develop, test and demonstrate the website and Tableau workbook.	A standard laptop/desktop capable of running a web browser and Tableau Public/Desktop; exact CPU/GPU not specified.
Hardware — Memory	Memory for browser, local project files and Tableau authoring.	Use available system RAM; exact capacity not specified.
Hardware — Storage	Stores source dataset files, website files and Tableau project artefacts.	Local storage sufficient for the project files; exact size not specified.
Software — BI Tool	Data connections, calculations, dashboards and story.	Tableau Desktop / Tableau Public, as used in the project.
Software — Front End	Website layout, styling and client-side behaviour.	HTML, CSS, Bootstrap and JavaScript.
Software — Development Environment	Edit files, manage source and publish website.	Code editor and browser; GitHub may be used if the site is hosted there.
Data	F1 structured records used for driver, team, race, tyre, podium, DNF and country analysis.	Provided CSV/text dataset files; exact row count and file size not specified.

Expected Deliverable

A working project website that presents the F1 Racing Analytics introduction, the published Tableau dashboard, and the driver leaderboard story with its Top N control, subject to Tableau Public embed availability.

Project ID field intentionally left blank. Resource specifications that were not provided have not been invented.